Dimensionality Reduction

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Ch. 6 of I2ML (Secs. 6.4, 6.6, and 6.12 – 6.13 excluded)

Outline

Introduction

Subset Selection

Principal Component Analysis

Linear Discriminant Analysis

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Why Dimensionality/Dimension Reduction?

- Whether for classification or regression problem, observation data that we believe are informative are taken as inputs and fed to the system for decision making.



- The number of inputs (input dimensionality of the feature) often affects the time and space complexity of the learning algorithm (either classifier or regressor):
  - Having less computation reduces time complexity.
  - Having fewer parameters reduces space complexity.
- Eliminating an input deemed unnecessary saves the cost of extracting/observing it.
- Simpler models are often more robust on small data sets.
- Simpler models are more interpretable, leading to simpler explanation.
- Data visualization in 2 or 3 dimensions facilitates the detection of structure and outliers.

Feature Selection vs. Extraction

- Two main methods for reducing dimensionality: feature selection and feature extraction.
- Feature selection:
  - Choosing $k < d$ important features and discarding the remaining $d - k$.
  - Subset selection algorithms (supervised methods)
- Feature extraction:
  - Projecting the original d dimensions to k ($< d$) new dimensions.
  - Unsupervised methods (without using output information):
    - Principal component analysis (PCA)
    - Factor analysis (FA)
    - Multidimensional scaling (MDS)
    - Canonical correlation analysis (CCA)
  - Supervised methods (using output information):
    - Linear discriminant analysis (LDA)
  - The linear methods above also have nonlinear extensions.
- These k features may be interpreted as hidden or latent factors that in combination generate the observed d features.

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Subset Selection

- Goal: find the best subset of features.
- The best subset contains the least number of dimensions that most contribute to accuracy with respect to a certain task (e.g., classication, regression, visualization).
- There are $2^d - 1$ nonempty subsets of d features.
- Unless d is small, the search space is typically huge, making it impossible to conduct an exhaustive search for the best subset.
- Heuristic algorithms are often used to obtain reasonable (suboptimal) solutions in reasonable (polynomial) time.
- Two conventional approaches:
  - Forward search
  - Backward search
- More recent approach: using sparsity-inducing regularizers such as ℓ_1-norm.

Sequential Forward Search

- Start with no features and add them one by one, at each step adding the one that decreases the error measure the most, until the error cannot be further decreased.
- The error E (e.g., misclassication error for classification, mean squared error for regression) should be measured on a validation set distinct from the training set.
- Algorithm skeleton:
  - Initialize feature set as empty set: $\mathcal{F} = \emptyset$
  - At each iteration:
    - For each available feature x_i , train the model and calculate the error $E(\mathcal{F} \cup \{x_i\})$ incurred on the validation set.
    - Find the best feature x_j: $j = \arg\min_i E(\mathcal{F} \cup \{x_i\})$
    - If $E(\mathcal{F} \cup \{x_j\}) < E(\mathcal{F})$ then add x_j to F and continue; else exit.
- To select k features from d, we need to train and test the model $d + (d-1) + (d-2) + \cdots + (d-k+1)$ times, which is of the order $O(d^2)$.
- No guarantee for optimal subset with greedy search.
- We can add multiple features at a time (requires more computation) or backtrack to check which previously added feature can be removed.

Sequential Backward Search

- Start with all features and do a similar process as forward search except by removing features one at a time.
- Algorithm skeleton:
  - Initialize feature set $\mathcal{F}$ with all features.
  - At each iteration:
    - For each feature $x_i \in \mathcal{F}$, train the model and calculate the error $E(\mathcal{F} \backslash \{x_i\})$ incurred on the validation set.
    - Find the best feature x_j: $j = \arg\min_i E(\mathcal{F} \backslash \{x_i\})$
    - If $E(\mathcal{F} \backslash \{x_i\}) < E(\mathcal{F})$ then remove x_j from $\mathcal{F}$ and continue; else exit.
- We can stop if removing a feature does not decrease the error.
  - For model complexity reduction, we may decide to remove a feature if its removal causes only a slight increase in error. (similar procedure may apply to forward search)
- To select k features from d, we need to train and test the model $d + (d-1) + (d-2) + \cdots + (k+1)$ times.
- Backward search is more computationally demanding than forward search:
  - Usually $k \ll d$
  - Training a model with more features is more costly.

Remarks on Feature Selection

- In applications like face recognition, feature selection is not a good method for dimensionality reduction because individual pixels by themselves do not carry much discriminative information; it is the combination of values of several pixels together that carry information about the face identity.
- Dimensionality reduction in such cases is done by feature extraction methods that we will discuss next.

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Principal Component Analysis

- Projection methods (feature extraction methods) aim to find a linear mapping from the d-dimensional input space ($\mathbf{x}$-space) to a k-dimensional space ($k \ll d$) ($\mathbf{z}$-space) with minimum information loss according to some criterion.
  - scalar projection of $\mathbf{x}$ on the direction of $\mathbf{w}$ s.t. $\|\mathbf{w}\| = 1$:

$$z = \mathbf{w}^T \mathbf{x}$$

- Principal component analysis (PCA) is one of the projection methods.
- The principal component is $\mathbf{w}_1$ such that the sample, after projection on to $\mathbf{w}_1$, is most spread out so that the difference between the sample points becomes most apparent and hence the criterion to be optimized is the variance.
- Finding the first principal component $\mathbf{w}_1$ s.t. the $\mathrm{Var}(z_1)$ is maximized:

$$\begin{aligned}\mathrm{Var}(z_1) &= \mathrm{Var}(\mathbf{w}_1^T \mathbf{x}) = \mathbb{E}[(\mathbf{w}_1^T \mathbf{x} - \mathbb{E}(\mathbf{w}_1^T \mathbf{x}))^2] = \mathbb{E}[(\mathbf{w}_1^T (\mathbf{x} - \boldsymbol{\mu}))^2] \\ &= \mathbb{E}[\mathbf{w}_1^T (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{w}_1] = \mathbf{w}_1^T \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \mathbf{w}_1 = \mathbf{w}_1^T \boldsymbol{\Sigma} \mathbf{w}_1\end{aligned}$$

where

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}], \qquad \boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{x}) = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$$

Optimization Problem for First Principal Component

- The optimization problem is given by

$$\begin{aligned} \underset{\mathbf{w}_1}{\text{maximize}} \quad & \mathbf{w}_1^T \boldsymbol{\Sigma} \mathbf{w}_1 \\ \text{subject to} \quad & \|\mathbf{w}_1\| = 1 \end{aligned}$$

  which is a constrained optimization problem and the Lagrangian is given by

$$\mathcal{L}(\mathbf{w}_1, \alpha) = -\mathbf{w}_1^T \boldsymbol{\Sigma} \mathbf{w}_1 + \alpha(\mathbf{w}_1^T \mathbf{w}_1 - 1)$$

  where α is the Lagrange multiplier.

- Taking the derivative of the Lagrangian w.r.t. $\mathbf{w}_1$ and setting it to $\mathbf{0}$, we get an eigenvalue equation for the (first) principal component $\mathbf{w}_1$:

$$\boldsymbol{\Sigma} \mathbf{w}_1^\star = \alpha^\star \mathbf{w}_1^\star$$

- Since

$$(\mathbf{w}_1^\star)^T \boldsymbol{\Sigma} \mathbf{w}_1^\star = \alpha^\star (\mathbf{w}_1^\star)^T \mathbf{w}_1^\star = \alpha^\star$$

  we choose the eigenvector corresponding to the largest eigenvalue λ_1 for the objective to be maximized.

Alternative Problem Formulation for First Principal Component

- The optimization problem for the first principal component can also be written as

$$\underset{\mathbf{w}_1}{\text{maximize}} \quad \frac{\mathbf{w}_1^T \boldsymbol{\Sigma} \mathbf{w}_1}{\mathbf{w}_1^T \mathbf{w}_1}$$

  which is to maximize the Rayleigh quotient.
- The same solution can be obtained.

Optimization Problem for Second Principal Component

- The second principal component $\mathbf{w}_2$ defines the projection $z_2 = \mathbf{w}_2^T \mathbf{x}$, which should maximize $\text{Var}(z_2) = \mathbf{w}_2^T \mathbf{\Sigma} \mathbf{w}_2$ with z_2 uncorrelated to z_1, i.e.,

$$\text{Cov}(z_1, z_2) = \mathbb{E}[(\mathbf{w}_1^T \mathbf{x} - \mathbf{w}_1^T \boldsymbol{\mu})(\mathbf{w}_2^T \mathbf{x} - \mathbf{w}_2^T \boldsymbol{\mu})] = \mathbf{w}_1^T \mathbf{\Sigma} \mathbf{w}_2 = \lambda_1 \mathbf{w}_2^T \mathbf{w}_1 = 0$$

- The optimization problem is

$$\begin{aligned} \underset{\mathbf{w}_2}{\text{maximize}} \quad & \mathbf{w}_2^T \mathbf{\Sigma} \mathbf{w}_2 \\ \text{subject to} \quad & \|\mathbf{w}_2\| = 1,\ \mathbf{w}_2^T \mathbf{w}_1 = 0 \end{aligned}$$

- The Lagrangian:

$$\mathcal{L}(\mathbf{w}_2, \alpha, \beta) = -\mathbf{w}_2^T \mathbf{\Sigma} \mathbf{w}_2 + \alpha(\mathbf{w}_2^T \mathbf{w}_2 - 1) + \beta(\mathbf{w}_2^T \mathbf{w}_1 - 0)$$

- Taking the derivative of the Lagrangian w.r.t. $\mathbf{w}_2$ and setting it to $\mathbf{0}$, we get

$$2\mathbf{\Sigma} \mathbf{w}_2^\star - 2\alpha^\star \mathbf{w}_2^\star - \beta^\star \mathbf{w}_1 = \mathbf{0}$$

- We can show that $\beta^\star = 0$ and hence have this eigenvalue equation $\mathbf{\Sigma} \mathbf{w}_2^\star = \alpha^\star \mathbf{w}_2^\star$, implying that $\mathbf{w}_2^\star$ is the eigenvector of $\mathbf{\Sigma}$ with the second largest eigenvalue.

Optimization Problem for Other Principal Components

- Similarly, we can show that the other dimensions are given by the eigenvectors of $\mathbf{\Sigma}$ with decreasing eigenvalues.

- The sample covariance $\mathbf{S}$ is symmetric, so, for two different eigenvalues, the eigenvectors are orthogonal.
  - If $\mathbf{S}$ is positive definite, then all its eigenvalues are positive.
  - If $\mathbf{S}$ is singular, then its rank, the effective dimensionality, is k with $k < d$ and λ_i, $i = k+1, \ldots, d$ are 0 (λ_i are sorted in descending order).
    - The k eigenvectors with nonzero eigenvalues are the dimensions of the reduced space.
- The first eigenvector (the one with the largest eigenvalue λ_1), $\mathbf{w}_1$, namely, the first principal component, explains the largest part of the variance; the second explains the second largest; and so on.

- We have discussed obtaining the principal components through a variance maximization formulation, which provides a statistical view for PCA.

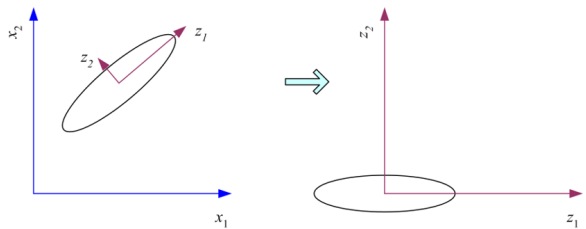
What PCA Does? – I

- Transformation of data:

$$\mathbf{z} = \mathbf{W}^T(\mathbf{x} - \mathbf{m})$$

where the k columns of $\mathbf{W} \in \mathbb{R}^{d \times k}$ are the k leading eigenvectors of the sample covariance $\mathbf{S}$, and $\mathbf{m}$ is the sample mean.

- PCA intuition: centering the data at the origin and rotating the axes:



If $\text{Var}(z_2)$ is too small, it can be ignored to reduce the dimensionality from 2 to 1.

- After the linear transformation, we get a k-dimensional space whose dimensions are the eigenvectors, and the variances over them are equal to the eigenvalues.

What PCA Does? – II

- The eigenvalue decomposition of the sample covariance $\mathbf{S}$ is given by

$$\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

where $\mathbf{\Lambda} \in \mathbb{R}^{d \times d}$ and $\mathbf{Q} \in \mathbb{R}^{d \times d}$ with $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ are the eigenvalue matrix and eigenvector matrix, respectively, and hence

$$\mathbf{Q}^T\mathbf{S}\mathbf{Q} = \mathbf{\Lambda}$$

- We have

$$\text{Cov}(\mathbf{z}) = \mathbf{W}^T\mathbf{S}\mathbf{W} = \mathbf{\Lambda}_k$$

which is a diagonal matrix.

- PCA intuition: find a matrix $\mathbf{W}$ s.t. the linear transformed data $\mathbf{z} = \mathbf{W}^T(\mathbf{x} - \mathbf{m})$ has diagonal covariance; that is, we would like to get uncorrelated z_i.
- PCA does not use output information and hence is a one-group procedure.

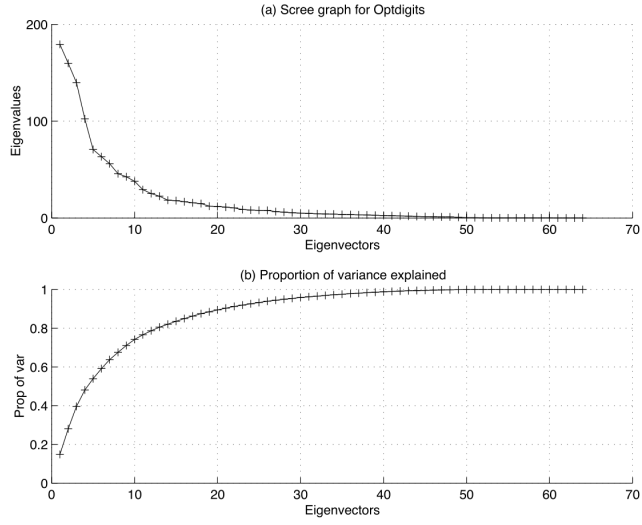
How to Choose k?

- Proportion of variance (PoV) explained (or cumulative explained variance):

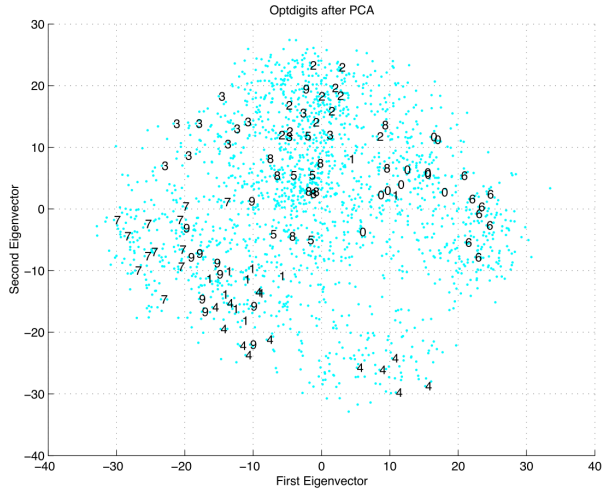
$$\frac{\lambda_1 + \lambda_2 + \cdots + \lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_k + \cdots + \lambda_d}$$

  where λ_d are sorted in descending order.
- Typically, stop at PoV > 0.9.
- Scree graph plotting PoV against k; stop at "elbow".

Scree Graph


(a) Scree graph for Optdigits
Eigenvalues
Eigenvectors
0
100
200
0
10
20
30
40
50
60
70
(b) Proportion of variance explained
Prop of var
Eigenvectors
0
0.2
0.4
0.6
0.8
1
0
10
20
30
40
50
60
70

PCA for Visualization: Scatterplot in Lower-Dimensional Space



An Alternative Equivalent Formulation for PCA

- Given the transformation of data $\mathbf{z} = \mathbf{W}^T(\mathbf{x} - \mathbf{m})$, where $\mathbf{W} \in \mathbb{R}^{d \times d}$ with $\mathbf{W}\mathbf{W}^T = \mathbf{I}$. We have

$$\mathbf{x} = \mathbf{m} + \mathbf{W}\mathbf{z}$$

- If $\mathbf{W} \in \mathbb{R}^{d \times k}$ with columns to be the principal components, the reconstruction of $\mathbf{x}^t$ from its representation in the lower-dimensional $\mathbf{z}$-space is

$$\hat{\mathbf{x}}^t = \mathbf{m} + \mathbf{W}\mathbf{z}^t \quad \text{or} \quad \mathbf{x}^t = \mathbf{m} + \mathbf{W}\mathbf{z}^t + \boldsymbol{\epsilon}^t$$

- It can be proved that among all orthogonal linear projections, PCA minimizes the reconstruction error (a geometric view of PCA), i.e.,

$$\begin{aligned} \underset{\mathbf{m},\mathbf{W},\{\mathbf{z}^t\}}{\text{minimize}} \quad & \frac{1}{2}\sum_{t=1}^{N} \|\mathbf{x}^t - (\mathbf{m} + \mathbf{W}\mathbf{z}^t)\|_2^2 = \frac{1}{2}\|\mathbf{X} - \mathbf{m}\mathbf{1}^T - \mathbf{W}\mathbf{Z}\|_F^2 \\ \text{subject to} \quad & \mathbf{W}^T\mathbf{W} = \mathbf{I} \end{aligned}$$

  - We can pre-subtract the sample mean from $\mathbf{x}^t$ or constrain $\sum_t \mathbf{z}^t = \mathbf{0}$ if we expect $\mathbf{m}$ to be the sample mean estimate of $\mathbf{x}^t$.

Probabilistic PCA

- PCA model is not a generative model, since the low-dimensional representation $\{\mathbf{z}^t\}$ and the error $\{\boldsymbol{\epsilon}^t\}$ are not treated as random variables. As a consequence, the PCA model cannot be used to generate new samples of the random variable $\mathbf{x}$.
- To address this issue, the probabilistic PCA (PPCA) assume that $\mathbf{z}$ and $\boldsymbol{\epsilon}$ are independent random variables with some pdfs, then it generates an $\mathbf{x}$ by

$$\mathbf{x} = \mathbf{m} + \mathbf{W}\mathbf{z} + \boldsymbol{\epsilon}$$

- Let the mean and covariance of $\mathbf{z}$ be denoted by $\boldsymbol{\mu}_z$ and $\boldsymbol{\Sigma}_z$ (commonly assuming $\boldsymbol{\Sigma}_z = \mathbf{I}_k$), respectively, and the mean and covariance of $\boldsymbol{\epsilon}$ be denoted by $\mathbf{0}$ and $\boldsymbol{\Sigma}_\epsilon$ (commonly assuming $\boldsymbol{\Sigma}_\epsilon = \psi^2 \mathbf{I}_d$). Then we have

$$\boldsymbol{\mu} = \mathbf{m} + \mathbf{W}\boldsymbol{\mu}_z \quad \text{and} \quad \boldsymbol{\Sigma} = \mathbf{W}\boldsymbol{\Sigma}_z\mathbf{W}^T + \boldsymbol{\Sigma}_\epsilon$$

- Then we can estimate $\mathbf{m}$, $\mathbf{W}$, $\boldsymbol{\mu}_z$, $\boldsymbol{\Sigma}_z$, and $\boldsymbol{\Sigma}_\epsilon$ from the estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ or directly from the sample $\{\mathbf{x}^t\}$ through, say, MLE.

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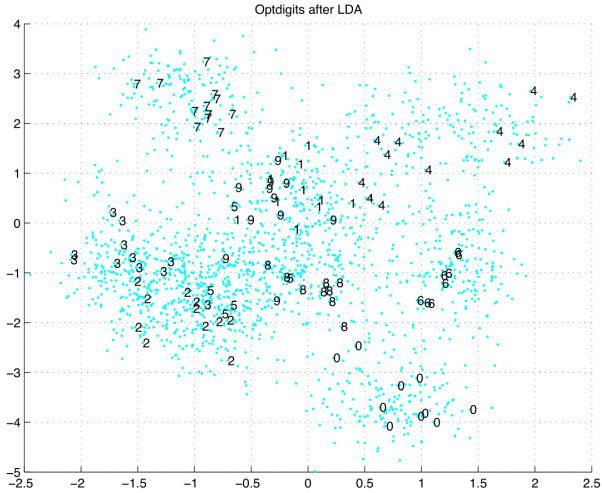
Principal Component Analysis

Linear Discriminant Analysis

Fisher Linear Discriminant Analysis (LDA)

- Unlike PCA, FA, and MDS, LDA is a supervised dimensionality reduction method.
- LDA is typically used with a classifier for classification problems.
- Goal: the classes are well-separated after projecting to a low-dimensional space by utilizing the label information (output information).

Example



2-Class Case

- Given sample $\mathcal{X} = \{(\mathbf{x}^t, r^t)\}$, where $r^t = 1$ if $\mathbf{x}^t \in C_1$ (class 1) and $r^t = 0$ if $\mathbf{x}^t \in C_2$ (class 2).
- Find vector $\mathbf{w}$ on which the data are projected such that the examples from C_1 and C_2 are as well separated as possible.
- Projection of $\mathbf{x}$ onto $\mathbf{w}$ (dimensionality reduced from d to 1):

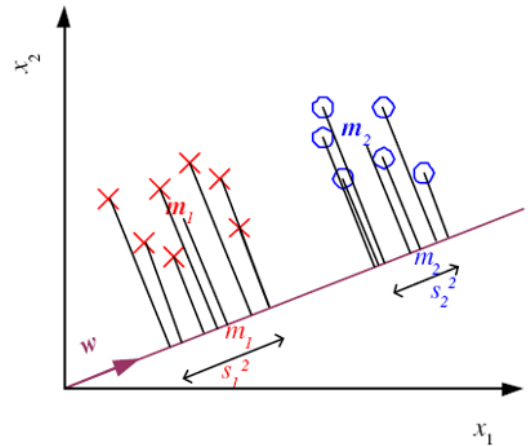
$$z = \mathbf{w}^T \mathbf{x}$$

- $\mathbf{m}_i \in \mathbb{R}^d$ and $m_i \in \mathbb{R}$ are sample means of C_i before and after projection:

$$m_1 = \frac{\sum_t \mathbf{w}^T \mathbf{x}^t r^t}{\sum_t r^t} = \mathbf{w}^T \mathbf{m}_1$$

$$m_2 = \frac{\sum_t \mathbf{w}^T \mathbf{x}^t (1 - r^t)}{\sum_t (1 - r^t)} = \mathbf{w}^T \mathbf{m}_2$$

Projection



Between-Class Scatter

- Between-class scatter (in a form of normalized sample variance/covariance matrix):

$$\begin{aligned}(m_1 - m_2)^2 =& (\mathbf{w}^T\mathbf{m}_1 - \mathbf{w}^T\mathbf{m}_2)^2 \\ =& \mathbf{w}^T(\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T\mathbf{w} \\ =& \mathbf{w}^T\mathbf{S}_B\mathbf{w}\end{aligned}$$

where the between-class scatter matrix

$$\begin{aligned}\mathbf{S}_B =& (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T \\ =& \frac{4N_1}{N_1 + N_2}(\mathbf{m}_1 - \mathbf{m})(\mathbf{m}_1 - \mathbf{m})^T + \frac{4N_2}{N_1 + N_2}(\mathbf{m}_2 - \mathbf{m})(\mathbf{m}_2 - \mathbf{m})^T\end{aligned}$$

where $\mathbf{m}$ is the mean of the class means, i.e., $\mathbf{m} = \frac{1}{2}(\mathbf{m}_1 + \mathbf{m}_2)$, and N_1 and N_2 are the # of instances in C_1 and C_2, respectively.

Within-Class Scatter

- Within-class scatter:

$$\begin{aligned} s_1^2 &= \sum_t (\mathbf{w}^T \mathbf{x}^t - m_1)^2 r^t \\ &= \sum_t \mathbf{w}^T (\mathbf{x}^t - \mathbf{m}_1)(\mathbf{x}^t - \mathbf{m}_1)^T \mathbf{w} r^t \\ &= \mathbf{w}^T \mathbf{S}_1 \mathbf{w} \end{aligned}$$

  where the within-class scatter matrix $\mathbf{S}_1 = \sum_t (\mathbf{x}^t - \mathbf{m}_1)(\mathbf{x}^t - \mathbf{m}_1)^T r^t$. Also,

$$s_2^2 = \mathbf{w}^T \mathbf{S}_2 \mathbf{w}$$

  with $\mathbf{S}_2 = \sum_t (\mathbf{x}^t - \mathbf{m}_2)(\mathbf{x}^t - \mathbf{m}_2)^T (1 - r^t)$.

- So, the total within-class scatter

$$s_1^2 + s_2^2 = \mathbf{w}^T \mathbf{S}_W \mathbf{w}$$

  where

$$\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2$$

Fisher's Linear Discriminant

- Fisher's linear discriminant refers to the vector $\mathbf{w}$ that maximizes the Fisher criterion (a generalized Rayleigh quotient):

$$\underset{\mathbf{w}}{\text{maximize}} \quad J(\mathbf{w}) = \frac{(m_1 - m_2)^2}{s_1^2 + s_2^2} = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

  which is equivalent to solve

$$\begin{aligned} \underset{\mathbf{w}}{\text{maximize}} \quad & \mathbf{w}^T \mathbf{S}_B \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{S}_W \mathbf{w} = 1 \end{aligned}$$

- We can prove the optimal solution satisfies the following generalized eigenvalue problem:

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}$$

  or, if $\mathbf{S}_W$ is nonsingular, an equivalent eigenvalue problem:

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}$$

2-Class Case

- For the 2-class case, we note that

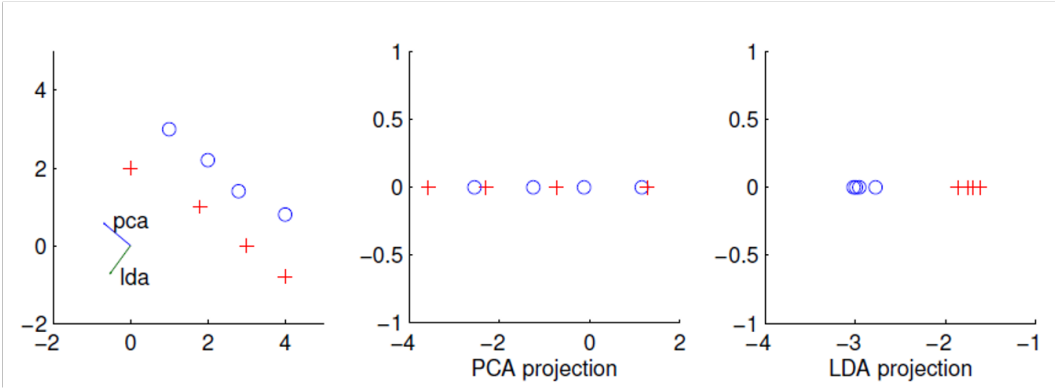
$$\mathbf{S}_B\mathbf{w} = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T\mathbf{w} = c(\mathbf{m}_1 - \mathbf{m}_2)$$

  for some constant c and hence $\mathbf{S}_B\mathbf{w}$ (also $\mathbf{S}_W\mathbf{w}$) is in the same direction of $\mathbf{m}_1 - \mathbf{m}_2$.
- So we get

$$\mathbf{w} = c\mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2) = c(\mathbf{S}_1 + \mathbf{S}_2)^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

  where c is some irrelevant constant factor.
- We have projected the samples from d dimensions to 1, i.e., dimensionality reduction, and any classification method can be used afterward.
- Recall that when $p(\mathbf{x}|C_i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$ (i.e., homoscedasticity), we have a linear discriminant where $\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, and we see that Fisher's linear discriminant is optimal if the classes are normally distributed. But Fisher's linear discriminant can be used even when the classes are not normal.

PCA vs. LDA



$K > 2$ Classes – I

- Find the matrix $\mathbf{W} \in \mathbb{R}^{d \times k}$ such that

$$\mathbf{z} = \mathbf{W}^T \mathbf{x} \in \mathbb{R}^k$$

- Within-class scatter matrix for C_i:

$$\mathbf{S}_i = \sum_t r_i^t (\mathbf{x}^t - \mathbf{m}_i)(\mathbf{x}^t - \mathbf{m}_i)^T$$

  where $r_i^t = 1$ if $\mathbf{x}^t \in C_i$ and 0 otherwise.

- Total within-class scatter matrix:

$$\mathbf{S}_W = \sum_{i=1}^K \mathbf{S}_i$$

$K > 2$ Classes – II

- Between-class scatter matrix:

$$\mathbf{S}_B = \sum_{i=1}^{K} N_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^T$$

  where $\mathbf{m}$ is the overall mean (i.e., mean of the class means) or sometimes chosen as the mean weighted by sample numbers and $N_i = \sum_t r_i^t$.

- We need to find the matrix $\mathbf{W}$ that maximizes

$$\underset{\mathbf{W}}{\text{maximize}} \quad J(\mathbf{W}) = \frac{\det(\mathbf{W}^T \mathbf{S}_B \mathbf{W})}{\det(\mathbf{W}^T \mathbf{S}_W \mathbf{W})}$$

  which corresponds to the eigenvectors of $\mathbf{S}_W^{-1}\mathbf{S}_B$ with the largest eigenvalues.

- Take $k \leq K - 1$: since $\mathbf{S}_B$ is the sum of K rank-1 matrices and only $K - 1$ of them are independent, $\mathbf{S}_B$ has a maximum rank of $K - 1$.